

Captured by the Words: Literal Attraction in Typed Decision Models Is Governed by Language, Not Capability, in Bangla Idioms

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Abstract

Decision models, non-autoregressive scorers that return a calibrated probability over caller-supplied options and emit no text, are increasingly deployed as routers, classifiers and guardrails. They generate nothing, so the usual explainability toolkit of rationales, chain-of-thought and token attribution does not apply, and idioms make token-level attribution ill-posed in principle: a non-compositional expression has no word whose removal is informative. We therefore audit three such models, an encoder scorer, a decoder scorer and a closed commercial API, entirely through *interventions on the option set*, using 10,361 expert-annotated Bangla idioms that ship with both a gold figurative gloss and a gold literal gloss. Our instrument holds the item, the instruction and k fixed and varies only the distractor family, which isolates one failure mode at a time and needs no new annotation. Replacing one distractor with an idiom’s own literal gloss captures all three models, and a full model $\times$ language matrix on 3,265 shared items shows the effect is governed by language rather than capability: literal attraction falls from 0.731, 0.918 and 0.304 in Bangla to 0.379, 0.241 and 0.047 in English. The commercial model’s apparent resistance to capture is a property of the language it was tested in. Capability does not order the failure either, as the strongest arm is the least captured while the middle arm is the most. Confidence inverts under the same shift, so confidence-gated selective prediction does worse than random gating, and on a semantics-free option list, where accuracy is pinned at chance by construction, the three arms still report 0.28 to 0.82 confidence. Option-order sensitivity, by contrast, separates the two scoring architectures rather than the three models: 83.7% against 10.7% of items change their chosen gloss across the 24 orderings of the same four options.

আবর্জনা abarjana

literally “garbage or waste” · figuratively “an unwanted person”

ASKED IN BANGLA

literal	আবর্জনা (garbage or waste)	0.85	✗
figurative	অবাঞ্ছিত ব্যক্তি	0.09	

ASKED IN ENGLISH

literal	garbage or waste	0.04	
figurative	Unwanted person	0.96	✓

Figure 1: **Literal capture is a property of the language, not the model.** The Bangla idiom *abarjana* means an unwanted person; its words mean garbage. Jev-1.13, the strongest of the three models we audit, is asked the same question twice with the same four options, once in Bangla and once in English. In Bangla it puts 0.85 on the literal reading and 0.09 on the gold; in English it puts 0.96 on the gold. Colour marks the option’s role rather than the model. Across 3,265 items the same swap moves literal attraction from 0.304 to 0.047 for this model, and from 0.731 to 0.379 and 0.918 to 0.241 for the two open ones (§5.1). Figures rendered outside $\LaTeX$ carry the original script; see Appendix D.

1 Introduction

A *decision model* is given a question and a list of candidate options and returns a probability distribution over exactly those options. It is not autoregressive, it does not sample, and it produces no text. This class has grown quickly as infrastructure for routing, intent classification, content gating and tool selection, precisely because it is cheap, bounded and returns a number a caller can threshold on. An ecosystem analysis of 2,170 public projects finds it already embedded as a reusable decision component across domains (Ling et al., 2026), and proposals now place these models at the adjudication points of autonomous security harnesses (dos Santos, 2026). The threshold is load-bearing.

That bounded interface is also an explainability problem. There is no rationale to inspect, no chain of thought to measure for faithfulness (Jacovi and Goldberg, 2020; Turpin et al., 2023), and in the commercial case no weights at all. What is left is the behavioural surface: the caller controls the question and the option list, and sees a probability vector. Any audit of this class has to be built out of that.

The obvious fallback, attributing the decision to input tokens (Ribeiro et al., 2016; Lundberg and Lee, 2017), is not merely unreliable here (Jain and Wallace, 2019; Adebayo et al., 2018; Hooker et al., 2019); for our domain it is ill-posed. An idiom is non-compositional by definition (Nunberg et al., 1994): the words of *kick the bucket* do not carry fractions of its meaning, so a Shapley value over them attributes a quantity that does not exist. The expression has to be intervened on as a unit, or not at all.

We take the option set as the intervention surface instead. Bangla *bāgdhārā* (idioms) are well suited to this because the expert resource we use ships, for each idiom, both a gold figurative meaning and a gold *literal* gloss: a ready-made, human-certified counterfactual distractor that no annotation effort of ours produced. Dropping that gloss into the option list turns a benchmark question into a targeted probe: does the model read the expression as an idiom, or as its words? Figure 1 shows one item answered both ways.

We contribute (i) a meaning-choice instrument for decision models in which the *distractor family* is the independent variable and everything else is held fixed, requiring no new annotation (§4); (ii) the Literal Attraction Rate, and with it a Literal Attraction Rate, measured as a full model $\times$ language matrix, which shows capture governed by language rather than capability (§5.1); (iii) evidence that confidence is anti-diagnostic under this shift across three different calibration philosophies, with confidence-gated selective prediction worse than random gating (§5.2); (iv) a clean architectural dissociation in option-order dependence, with a methodological consequence for anyone evaluating this class (§5.4); and (v) a semantics-free control, the one condition that runs identically on all three arms, bounding what each model reports when the options provably carry no information (§5.3).

2 Related Work

The decision-model class. The evaluation literature is still forming, and reports competitive quality at far lower cost with no independent accuracy advantage from the typed readout itself (Tang and Zheng, 2026; Rao and Callison-Burch, 2026; Ibrahim and Zaki, 2026; Guo et al., 2026; Yu and Yao, 2026). Two strands bear on ours. Ibrahim and Zaki (2026) report decision-model *calibration exceeding most LLM baselines*, and Li et al. (2026) build the confidence-gated cascade that follows from it; both depend on the returned confidence being diagnostic, and §5.2–§5.3 give conditions under which it is not. Concurrently Sun et al. (2026) find the decision follows an option’s *name* rather than its bound definition, and Xu (2026) flip 61.4% of correct decisions with benign context. Appendix P sets out how each relates to our design.

Idioms in NLP. Idiom resources and tasks are well established (Haagsma et al., 2020; Tanyar Madabushi et al., 2022; Chakrabarty et al., 2022), including for Bangla (Sakhawat et al., 2026), and the difficulty of figurative language for distributional models is long-noted (Shwartz and Dagan, 2019). Psycholinguistics frames the phenomenon as competition between a literal and a figurative route whose resolution depends on decomposability and context (Cacciari and Tabossi, 1988; Gibbs et al., 1989; Titone and Connine, 1999), and Oh et al. (2026) give mechanistic evidence that the competition is real inside a transformer: causal tracing recovers two pathways carrying the figurative and literal readings simultaneously, with early layers promoting the figurative one and a direct route keeping the literal one available.

Our D-LIT intervention is the behavioural counterpart of that picture rather than a replication of it: they can ask *how* the literal route stays available inside a model they can open, we can only ask *how often* it wins in one that emits no text and, in one arm, exposes no weights (Appendix P).

Evaluating option scorers. Option-order sensitivity and selection bias are documented for prompted generative LMs (Pezeshkpour and Hruschka, 2024; Wei et al., 2024; Zheng et al., 2024; Zhao et al., 2021). We are not aware of a comparison of option-marker against independent-option scoring in small non-autoregressive models over the full permutation set, which is what §5.4 provides.

Calibration and selective prediction. Proper scoring and calibration measurement are standard (Brier, 1950; Gneiting and Raftery, 2007; Guo et al., 2017; Nixon et al., 2019; Kumar et al., 2019), as is the observation that calibration degrades under distribution shift (Ovadia et al., 2019; Jiang et al., 2021). Selective prediction gives the operational test (Geifman and El-Yaniv, 2017, 2019; Traub et al., 2024). We report the combination that matters for deployment: accuracy *decreasing* with confidence, and a confidence gate underperforming a random one.

Behavioural testing. Our design follows behavioural checklists, contrast sets and counterfactual minimal pairs (Ribeiro et al., 2020; Gardner et al., 2020; Kaushik et al., 2020; Naik et al., 2018; Jia and Liang, 2017); the contribution is carrying that tradition into a model class with no text output, where it is one of the few instruments left.

Low-resource evaluation. Bangla has strong encoders and corpora (Bhattacharjee et al., 2022; Kakwani et al., 2020), and cross-lingual benchmarks establish the transfer gap (Hu et al., 2020; Ruder et al., 2021; Adelani et al., 2021); §5.7 localises our failure to the Bangla side rather than to figurativeness.

3 Dataset

Source corpus. We use *Bangla Bagdhara* (Sakhawat et al., 2026), a corpus of 10,361 Bangla (Bengali) idioms annotated under a 19-field schema by expert consensus.¹ Each row carries the idiom, a gold figurative meaning in Bangla, an expert literal gloss, semantic tags, alternative surface forms certified as meaning-preserving, and a gold carrier sentence. We add no annotation; every intervention below is assembled from fields that already exist.

Its own benchmark on the same underlying task reports no model among 30 multilingual and instruction-tuned LLMs above 50% accuracy, against 83.4% for human annotators. Option construction differs, so those figures are not directly comparable to ours, but they locate the difficulty in the material rather than in our instrument.

4 Methodology

Figure 2 sets out the instrument end to end: what the corpus supplies, what each condition changes,

¹Two rows are lost to parsing, leaving 10,359; of these 8,876 carry every field the instrument requires.

and what comes back.

4.1 The meaning-choice task

Given an idiom, choose its gold figurative meaning from $k = 4$ candidates. Chance is 0.250 throughout. The **distractor family is the independent variable**; the item, the instruction, k , and per-item option-order randomisation (seeded by item id) are held fixed across families, so a paired test across families is a test on byte-identical questions.

4.2 Distractor families and measures

The five families are D-RAND (other idioms’ glosses at random), D-SURF (glosses sharing surface tokens with the gold), D-SEM (glosses from idioms sharing semantic tags), D-LIT (one distractor replaced by the idiom’s *own* expert literal gloss) and D-HARD (all three pressures at once). Appendix C gives each sampling rule, and Figure 7 one complete item in the original script.

For the families containing a literal gloss, with ℓ_i the literal option and g_i the gold, the Literal Attraction Rate is

$$\text{LAR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\arg \max_o p_i(o) = \ell_i \right].$$

LAR is not $1 - \text{accuracy}$: it isolates the errors that are specifically a literal reading, and has no floor, since a model that never reads literally scores 0 while still being wrong.

4.3 Interventions and statistics

Seven interventions vary one thing each: the distractor family (E1), the idiom’s surface form (E2), the idiom as a unit (E3), the task language (E4), nothing (E5, derived from E1), the option text (E6) and the option order (E7). Appendices C and A give what each holds fixed and the typed question they share.

All intervals are 95% percentile bootstrap over items (2,000 resamples); paired comparisons use McNemar’s exact test; ECE uses 15 equal-width bins and AURC is reported against a random-gate baseline. The two open arms were built independently from a written contract, and on all 4,440 overlapping rows the gold key, gold position, idiom string and literal-distractor key agree at 1.000, so every paired test is on identical questions (Appendix Q).

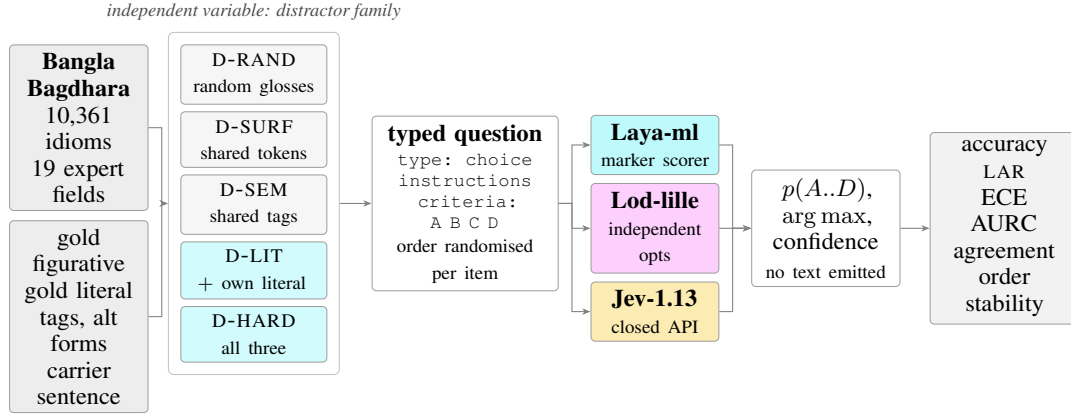


Figure 2: The instrument in one view. The corpus supplies both a gold figurative gloss and a gold *literal* gloss for each idiom, so the literal-bearing families (D-LIT and D-HARD) are assembled from expert annotation rather than written by us. Only the distractor family changes across conditions: the idiom, the instruction, $k = 4$ and the per-item order seed are held fixed, so any accuracy difference between two families on the same item is caused by the options alone. All three models receive the identical typed question (Laya-ml, Lod-lille, Jev-1.13) and return a probability over the four option keys and nothing else, which is the surface the audit works with.

	Laya-ml	Lod-lille	Jev-1.13
backbone	mmBERT	Qwen3-0.6B	closed
scoring	marker	independent	unknown
params	322M	0.6B	n/a
items	8,876	888	8,761
latency	27.2 ms	86.1 ms	API
weights	open	open	closed

Table 1: The three arms. Cell tint identifies the model throughout the paper: cyan Laya-ml, magenta Lod-lille, yellow Jev-1.13. Lod-lille ran a fixed 10% random subset; Appendix N bounds the resulting drift at 0.016.

Distractors	Laya	Lod	Δ
random	0.367	0.484	+0.133***
surface	0.338	0.412	+0.073***
tag	0.333	0.446	+0.120***
+ literal	0.151	0.066	-0.070***
hard	0.144	0.063	-0.086***
LAR (lit.)	0.644	0.885	
LAR (hard)	0.631	0.855	
mean $p(\text{lit.})$	0.464	0.726	
mean $p(\text{gold})$	0.195	0.128	

Table 2: The distractor ladder and literal attraction, McNemar on the 888 shared items (** $p < 0.001$). Bold marks the higher value per row, which is the better arm on the accuracy rows and the more captured arm on the LAR and $p(\text{lit.})$ rows. The sign of Δ flips at the horizontal rule: capability on the surrounding task does not protect against literal capture, and here is associated with more of it.

4.4 Models

Three models span the design space (Table 1). **Laya-ml** is a bidirectional encoder scoring all options together from marker positions. **Lod-lille** is a decoder with independent-option scoring, each option attending only to the shared state, the question and itself from the same start position, which makes it order-independent by construction in fp32. **Jev-1.13** is a closed commercial model behind a decisions API, documented as English-only; we know its interface and nothing else. Appendix A gives the call each one receives.

5 Analysis

5.1 Literal capture, and what it tracks

Table 2 and Figure 3 give the ladder for the two open arms on the 888 shared items. Lod-lille is the stronger model on every non-literal condition, by up to +0.133. Insert the idiom’s own literal gloss and the ordering *reverses*: Lod-lille falls to 0.066, less than half of Laya-ml’s already-below-chance 0.151, and barely a quarter of chance. The same reversal holds for D-HARD.

Within this pair, then, LAR measures something accuracy does not: a model can be selected as “better” on a conventional benchmark while being worse at what that benchmark proxies for. Laya-ml and Lod-lille answer byte-identical questions in the same language, so the crossover between them is a paired result. The error is not diffuse: the decomposition under D-LIT is gold / literal / other = 0.151 / 0.644 / 0.207 for Laya-ml and 0.066 / 0.885 / 0.048 for Lod-lille, and on the clearest items Laya-ml places $p = 1.000$ on the literal gloss.

Capture tracks the language, not capability.

The arms above ran the literal condition in Bangla; the commercial arm’s headline LAR = 0.047 is *English*, so the two are not comparable as they stand. We therefore ran the full model $\times$ language matrix

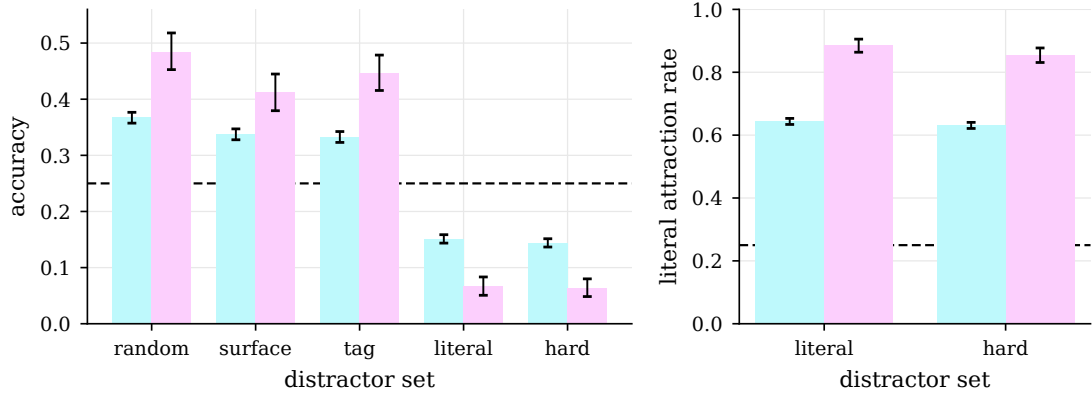


Figure 3: The distractor ladder (left) and literal attraction rate (right) for the two open arms. ■ Laya-ml ($n = 8,876$), ■ Lod-lille ($n = 888$); black dashed line is chance (0.250), error bars 95% bootstrap CIs. Surface- and tag-matched distractors cost a few points. Inserting the idiom’s own literal gloss (D-LIT) collapses both models far below chance, and collapses the *stronger* model harder: Lod-lille leads Laya-ml by $+0.133$ on random distractors and trails it by -0.070 once the literal gloss is present. On the right, both models prefer the literal reading to the figurative one on the majority of items.

on the 3,265-item shared pool, every cell built by one function from one seed (Table 12, Figure 4). The English cell for Jev-1.13 reproduces its published value at 0.0472 [0.0401, 0.0545], which is the check that the two constructions measure the same thing.

Two things follow. First, **every arm is far more captured in Bangla than in English on identical items**: Laya-ml $0.731 \rightarrow 0.379$, Lod-lille $0.918 \rightarrow 0.241$, Jev-1.13 $0.304 \rightarrow 0.047$, all paired McNemar $p < 10^{-100}$. The commercial model’s apparent resistance to literal capture was a property of the language it was tested in, not of the model. Second, **capture is not monotone in capability**. Ordered by Bangla D-RAND accuracy the arms run Laya-ml 0.367, Lod-lille 0.484, Jev-1.13 0.736; ordered by Bangla LAR they run Jev-1.13 0.304, Laya-ml 0.731, Lod-lille 0.918. Lod-lille is better than Laya-ml and more captured, while Jev-1.13 is better than both and less captured, so capability neither predicts capture nor protects against it.

The two open arms fall below chance on this condition (Laya-ml 0.127, Lod-lille 0.044); Jev-1.13 does not (0.536), but it loses 0.390 of accuracy relative to its own English run on the same items. One caveat stays: this pool is not a neutral sample, since items carrying a usable English literal gloss are the more literally-attracting ones.

The one matched three-way comparison. Exactly one condition is identical across all three arms: Bangla idiom, Bangla gloss, random distractors. Figure 11 and Table 3 report it. The commercial model is far ahead (0.736 against 0.367 and 0.484),

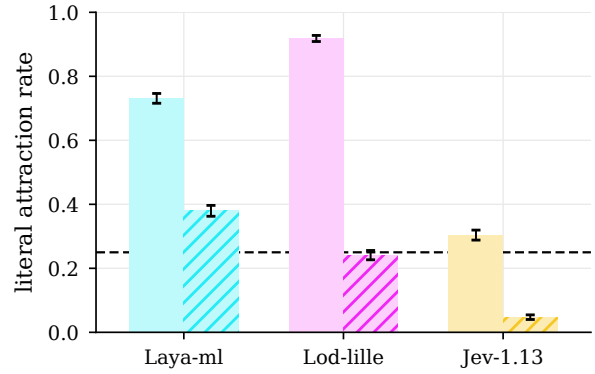


Figure 4: Literal attraction by model and task language, all six cells measured on the same 3,265 items ($n = 3,265$ each, error bars 95% bootstrap CIs, black dashed line chance). ■ Laya-ml, ■ Lod-lille, ■ Jev-1.13; solid bars are the Bangla condition, hatched bars the English one. Every arm is far more captured in Bangla than in English on identical items, and the Bangla ordering does not follow capability: Jev-1.13 is the most accurate of the three and the least captured, while Lod-lille is more accurate than Laya-ml and the most captured. Jev-1.13’s English value reproduces the figure reported from its original run, 0.047. Full numbers in Table 12.

Model	n	Accuracy	95% CI
Laya-ml	8,876	0.367	[.357, .377]
Lod-lille	888	0.484	[.453, .518]
Jev-1.13	8,761	0.736	[.727, .745]

Table 3: The only condition identical across all three arms: Bangla idiom $\rightarrow$ Bangla gloss, random distractors. All pairwise McNemar tests on the overlapping items are significant: Jev–Laya $+0.369$, Jev–Lod $+0.259$, Lod–Laya $+0.133$ (all $p < 10^{-9}$).

which is the honest capability ordering, and the one against which the calibration results of §5.2 and §5.3 should be read.

Outcome	Laya	Lod	Jev
correct	0.432	0.453	0.935
captured	0.596	0.787	0.737
other error	0.421	0.399	0.746
ECE (D-RAND)	0.071	0.073	0.005
ECE (D-LIT)	0.385	0.679	0.014
AURC (conf.)	0.924	0.989	0.013
AURC (random)	0.849	0.933	0.069

Table 4: Confidence by outcome on the literal-bearing families, and calibration. Bold marks, per model, whether the model is more confident when correct or when captured. Both open arms are more confident when captured; their confidence-gated AURC is *worse* than the random-gate baseline in the grey row, which inverts selective prediction. Jev-1.13’s English condition does not show the inversion.

5.2 Confidence is anti-diagnostic

Table 4 gives the mean probability assigned to the chosen option, split by outcome, on the literal-bearing families. Both open models are *more confident when captured than when correct*, Lod-lille extremely so (0.787 against 0.453). The reliability curves in Figure 5 slope *downward* under D-LIT in both.

Selective prediction therefore inverts: restricting Laya-ml’s D-LIT to its most confident 5% gives 0.007 accuracy against 0.151 at full coverage, and Lod-lille answers every item in its most confident 35% incorrectly (Appendix F). This is not an artifact of one calibration philosophy. Laya-ml ships fitted per-bucket temperatures, Lod-lille ships none by design, and Lod-lille’s *separate confidence head*, meant to estimate whether the answer is in the option list at all, shows the same inversion (0.631 on captured items against 0.295 on correct ones).

Jev-1.13 is the interesting exception, and it is only an exception in one direction. Its reliability in English is close to ideal on the two families in Table 4 (ECE 0.005 and 0.014, rising to 0.047 on D-HARD), and its confidence-gated AURC beats random by an order of magnitude. That makes §5.3 harder to dismiss, not easier.

This does not contradict Ibrahim and Zaki (2026), who find decision-model calibration better than most LLM baselines; we reproduce that for the commercial arm on its own language. The narrower claim is that in-distribution calibration need not survive a targeted distractor shift, and that it fails in the costly direction: a confidence-gated cascade (Li et al., 2026) accepts the high-confidence tail without review, and on the literal-bearing families

that tail is where the open arms are *least* accurate.

5.3 A semantics-free control, on all three arms

The O4 ablation replaces every option’s text with a bare label. No semantic signal remains, so accuracy is pinned at chance by construction and whatever confidence survives is overconfidence by definition, with no modelling assumption needed to read it. This is the only condition in the paper that runs identically on all three arms: it requires no gloss, no literal annotation and no shared language, so the language confound that scopes §5.1–§5.2 to the two open arms does not arise here. Figure 6 reports it.

The ordering reverses against Table 4. The best-calibrated model under real options is the worst when the options carry nothing: Jev-1.13 reports 0.815 mean confidence at 0.254 accuracy, an ECE of 0.561. Lod-lille, the weakest of the three on accuracy, is the only one whose reported confidence collapses appropriately (0.282). We read this as the most portable finding in the paper, because it is the one that needs no annotation and no matched language to reproduce: a calibration number measured on in-distribution options says little about what a model will report when the option list is degenerate, and a caller thresholding on a returned probability is exposed to that case. Confidence-based acceptance is the mechanism of the cascade designs proposed for this class (Li et al., 2026; Ibrahim and Zaki, 2026), and a degenerate option list is an ordinary consequence of an upstream bug or an empty retrieval rather than an adversarial construction.

A side-check comes with it: McNemar between the two open arms under O4 is not significant ($p = 0.76$), so the instrument’s null behaves identically in both.

5.4 Option order is architecture-specific

We run every item under all $4! = 24$ orderings of a fixed option set and record the identity of the chosen gloss, recovered as $\text{perm}[\text{slot}(\hat{y})]$.²

Table 5 and Figure 15 give a clean dissociation: a $7.8\times$ difference in answer stability and a $12\times$ difference in probability stability, tracking the scoring architecture rather than model size, backbone or capability. Lod-lille’s design claim, that each option

²An earlier version of this analysis counted distinct option keys (slot letters) and reported near-total instability for Laya-ml. That metric is an artifact: a key necessarily changes when the same gloss moves slots. All numbers here use the corrected gloss-identity metric; Laya-ml’s figure drops from 1.000 to 0.837 and the finding stands.

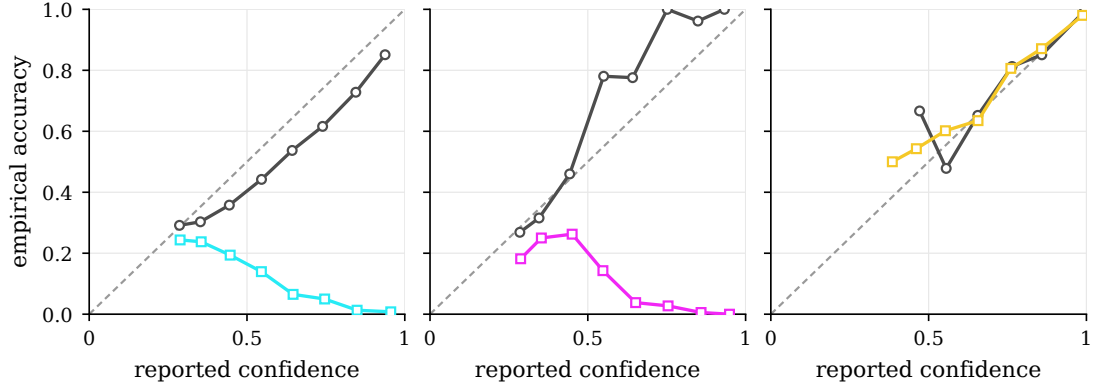


Figure 5: Reliability, 15 equal-width bins. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. ■ circles are the random-distractor condition; squares in each panel’s own colour are the +literal condition. The grey dashed diagonal is perfect calibration. Under D-LIT the two open models’ curves bend *downward*, higher stated confidence buying lower accuracy, while Jev-1.13, running in English, stays on the diagonal. Reading the left two panels: the model is most wrong exactly where it reports being most sure.

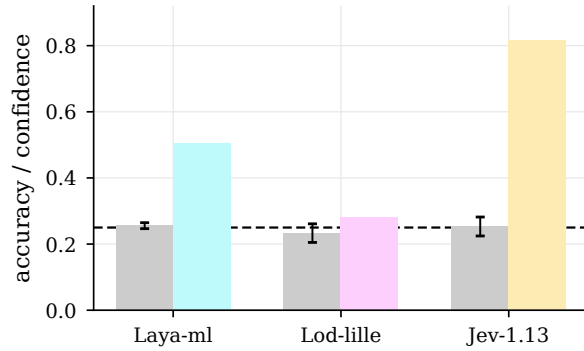


Figure 6: The semantics-free control. Per model, the ■ bar is accuracy (with 95% CI) and the coloured bar is mean reported confidence (■ Laya-ml, ■ Lod-lille, ■ Jev-1.13); black dashed line is chance. Because the options carry no information, accuracy is at chance by construction and the gap between the two bars is pure overconfidence: 0.250 for Laya-ml, 0.049 for Lod-lille, and 0.561 for Jev-1.13, which reports 0.815 confidence while answering at 0.254. Jev-1.13’s condition is English.

	Laya	Lod	Jev
answer changes	0.837	0.107	0.133
correctness varies	0.686	0.070	0.120
distinct glosses	2.83	1.14	1.15
perfectly stable	0.163	0.893	0.867
$p(\text{gold})$ range	0.346	0.029	0.128
selection spread	0.226	0.052	0.072

Table 5: Option-order sensitivity over all 24 orderings of the same four options ($n = 992 / 299 / 300$ items). Bold marks the least order-stable arm per row. The marker-based scorer changes its chosen gloss on 83.7% of items; the independent-option scorer, which is order-independent by construction in fp32, changes it on 10.7% and very nearly delivers what its design claims.

attends only to the shared state, its question and itself from the same start position, is very nearly borne out. Laya-ml’s option-marker scorer is not,

confirming an upstream issue report at scale.

The residual positional bias follows (Figure 14). With the gold placed uniformly, Laya-ml selects slot B 37.2% of the time against 14.6% for slot A, and its accuracy is 0.152 when the gold sits in A against 0.403 in B, a $2.6\times$ swing from position alone.

Methodological consequence. Any evaluation of this class that fixes option order is partly measuring position. All results here randomise order per item, seeded by item id, and are averages over position rather than measurements at one of them; on the evidence here we would treat randomising order as a default for this class rather than as a robustness check.

5.5 Options are read as a bag of words

The O-series ablates the option text while holding the idiom fixed (Figure 13; Figure 8 shows the forms applied to real Bangla). Truncating each option to six words and *scrambling the words inside it* cost almost nothing: Laya-ml retains 0.967 and 0.925 of its signal, Lod-lille 0.909 and 0.948, Jev-1.13 is flat to within noise. What all three do on the option side is close to order-free lexical matching. The drop to the grey semantics-free bar is the only large one, which certifies the instrument: the task measures option semantics, as a bag.

This converges with Sun et al. (2026), who permute which option name is bound to which definition and find the decision follows the name: within an option, these models read lexical identity rather than composed meaning.

5.6 Neither open model treats the idiom as a unit

E3 perturbs the idiom while holding the option list fixed (Figure 12). This is the experiment that replaces token attribution: since an idiom has no word-level meaning to attribute to, we instead ask whether the model behaves as though it stores the expression as a unit. A model that did would collapse when the unit is broken.

Neither does. Word-shuffling leaves Laya-ml’s accuracy statistically unchanged ($0.158 \rightarrow 0.168$) and reversal helps slightly (0.180); Lod-lille moves within noise. Agreement with the intact answer under shuffling is 0.742 and 0.917 , so Lod-lille is more *stable* but no more *compositional*. Figure 8 shows the perturbed strings.

The one manipulation that clearly helps both is embedding the idiom in its gold carrier sentence, $+0.080$ and $+0.117$: context, not the idiom string, carries the usable signal. These models are not retrieving a stored figurative entry but matching whatever lexical material is available. That cuts both ways, since Xu (2026) flip 61.4% of initially correct decisions with benign context, and nothing in the interface distinguishes the two cases.

5.7 The failure localises to Bangla, in every arm

Holding the model fixed and swapping only the task language (Figure 10) moves every arm by a large margin: Laya-ml $+0.246$, Lod-lille $+0.380$, Jev-1.13 $+0.233$ (all McNemar $p < 0.001$). Lod-lille reaches 0.864 in English against 0.484 in Bangla, so the figurative concepts are largely present; it simply cannot reach them through Bangla. Its documentation states that training evaluation was English-only, and this is what that looks like from outside the model.

Laya-ml’s control arm supports this reading: its English-only checkpoint run on Bengali script sits at 0.283 , barely above chance, which reproduces a documented failure mode and makes it unlikely that the Bangla result is an artifact of the scorer rather than of the language.

Jev-1.13 complicates the simple reading and then confirms it. It reaches 0.736 on the Bangla D-RAND task (Table 3) against 0.367 and 0.484 for the two models documented as multilingual, so a model with no multilingual claim outscores both: the Bangla gap is reachable, not intrinsic to the script. But §5.1 shows the same model’s literal

attraction rising from 0.047 in English to 0.304 in Bangla on identical items. Being better at Bangla is not the same as being robust in it, and on this evidence the language penalty reaches every arm we tested, including the one that was never advertised as multilingual.

6 Discussion

The matrix in §5.1 is the result we would defend hardest, because it separates two explanations that are easy to conflate. Read only in English, the commercial arm looks immune to literal capture; read in Bangla on the same items it is captured at six times the rate. What a decision model does with a non-compositional expression is therefore not a fixed property of the model, and a benchmark run in one language will not tell you how it behaves in another. Capability does not order the failure either: the strongest arm is the least captured, the middle arm the most.

The other results sit at different scopes and are worth keeping apart. The semantics-free control runs identically on all three arms and needs no annotation or shared language, so it is the one we would most expect to transfer. The confidence inversion is a paired result on the two open arms. Option-order dependence tracks the scoring architecture rather than the model, and is the one place our audit confirms a design claim rather than denting one.

For practitioners we would draw three cautions rather than recommendations: a returned probability is worth little without a measurement of what the model returns on a degenerate option list; a benchmark score is not by itself evidence against literal capture, since on our two open arms the two move in opposite directions; and an evaluation that fixes option order is partly reporting position.

This is a boundary condition rather than a rebuttal: the cost and latency advantages are established (Tang and Zheng, 2026; Rao and Callison-Burch, 2026), and what we add is that the property the cascade designs depend on degrades first and silently. The typed interface guarantees a well-formed answer and nothing about whether it was arrived at for a reason, so each of these failures is invisible to a caller who only checks that the output validates.

7 Conclusion

We audited three decision models on Bangla idioms using only interventions on the option set, a

method that needs no annotation and stays available when a model emits no text and its weights are closed. Replacing one distractor with an idiom’s own expert literal gloss captures all three, and running the full model $\times$ language matrix on a shared 3,265-item pool shows what the failure tracks: literal attraction falls from 0.731, 0.918 and 0.304 in Bangla to 0.379, 0.241 and 0.047 in English, so the commercial arm’s apparent resistance belongs to the language it was tested in rather than to the model. Capability does not order the failure, since the strongest arm is the least captured and the middle arm the most. Confidence inverts under the same shift, and with the options stripped of semantics, where accuracy is pinned at chance, the arms still report between 0.28 and 0.82 confidence. Option-order sensitivity separates the two scoring architectures rather than the three models. The instrument costs no annotation and, for the commercial arm, about eleven cents to run; we release it so the same audit can be pointed at other models and other languages.

Limitations

All three models are evaluated zero-shot, and this is the limitation we would weigh most heavily after the one below. Decision models are trained on a particular set of decision tasks, and the open arms’ own documentation reports that base checkpoints are weak on tasks outside that set. Some part of what we measure may therefore be out-of-distribution task transfer rather than literal capture specifically. The O4 control bounds this from one side, since it shows the instrument’s null behaving identically in both open arms, and the D-RAND to D-LIT contrast is within-model and within-item, which removes any fixed task-transfer offset. Neither argument rules the confound out. A single lightly fine-tuned arm would settle it, and we have not run one.

Laya-ml’s run had its per-bucket temperatures unset, so its ECE figures are raw. Lod-lille ships no temperature by design, so both open arms are effectively untempered and comparable on that axis; Jev-1.13’s internal calibration is unknown to us.

Lod-lille ran a fixed 10% subset (888 items), so its intervals are roughly ± 3 points and its stratified cells are thin. Appendix N bounds the subset bias at 0.016 and shows it changes no sign or ordering. E3 on the Lod-lille arm is 180 items and several of its perturbation contrasts are not individually

significant. E7 covers 992 / 299 / 300 items rather than the full sets.

The matrix in §5.1 runs on the 3,265 items that carry a usable English literal gloss, which is not a neutral sample: for Laya-ml, in-pool LAR is higher than out-of-pool while D-RAND accuracy is unchanged, so the absolute rates there are higher than they would be corpus wide. The language comparison is paired within item and within model, so that selection affects the levels rather than the direction. The matrix also covers the literal condition only; we did not run the other distractor families in both languages. The confidence results in §5.2 remain two-arm, since the commercial arm’s reliability was measured on its English run.

More broadly, this is one corpus, one language, two small open models and one commercial endpoint. The semantics-free control is the only result here that we would expect to transfer as stated; the rest should be read as findings about these systems on this material, and as an instrument other people can point at theirs.

Distractors are drawn from the same corpus, so difficulty is corpus-relative rather than absolute. Jev-1.13 is a commercial endpoint and may change without notice; our results are a snapshot, and we report the exact call count and cost in Appendix Q so the snapshot is at least auditable.

Finally, a typesetting note. pdfL^AT_EX cannot shape Bengali, so the plotted figures name their categories in Latin script. The stimuli themselves are shown in the original script in Figures 7, 9 and 8, which are rendered outside L^AT_EX by a shaping engine and embedded as vector; transliterations in the running text follow a loose convention and are for pronunciation only.

Ethics Statement

The corpus is a published, expert-annotated resource of idioms (Sakhawat et al., 2026); it contains no personal data and no content about identifiable individuals. Our interventions rewrite option lists and perturb idiom strings, and produce no new annotation and no generated text.

The work audits two openly released models and one commercial API. We report failure modes in all three, including the vendor’s. We have taken care to separate what is measured from what is inferred. An earlier version of this work reported the commercial arm’s low literal attraction without running it in Bangla; completing the matrix showed that

result was a property of the test language, and §5.1 now reports the corrected picture rather than the flattering one. §5.1 also states the pool selection effect against our own cleanest number. Commercial endpoints are mutable; readers should treat the Jev-1.13 results as dated rather than as a standing property of the product.

Bangla is under-served relative to its speaker population, and we think an audit that locates a failure in the language rather than in the task is the more useful kind to publish. The instrument, the per-item outputs and the analysis code are released; Appendix Q is the run record they correspond to.

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A The typed question

Every call in this paper sends one typed `choice` question and receives one probability vector. Nothing else is exchanged: there is no prompt template, no system message, no sampling parameter and no generated text. This is the whole interface, which is why the option set is the only intervention surface available.

Request. For the commercial arm the body posted to `/api/alpha/decisions` is

```
{"model": "typesafe/jev-1.13",
"state": <idiom, or carrier
sentence>,
"questions": {"q": {
"type": "choice",
"instructions": <one sentence,
Table 6>,
"criteria": {"A": <gloss>,
"B": <gloss>,
"C": <gloss>, "D": <gloss>}}}}
```

The two open arms are called locally with the same four fields: the idiom as state, the instruction string, and the four option texts keyed A–D. The option order is shuffled per item by a generator seeded with the item id and the experiment seed, so the same item receives the same permutation in every arm and every family.

Response.

```
{"type": "choice", "choice":
"B", "confidence": 0.75,
"probabilities": {"A": 0.00,
"B": 0.84,
"C": 0.16, "D": 0.00}}
```

We assert on every call that the returned keys are a subset of $\{A, B, C, D\}$ and that the probabilities sum to 1 within 10^{-2} ; no call in any run violated either.

Two confidence signals, and which one we use.

Note that `confidence` is not the maximum probability: in the example above it is 0.75 while $\max_o p(o) = 0.84$. It is a separate head whose documented job is to estimate whether the correct

answer is present in the option list at all. Lod-lille ships the same arrangement. We log it as `conf_head` and report it where it matters (§5.2), but every cross-model number in this paper uses the maximum probability, because it is the only confidence signal all three arms define identically. Where the two disagree the head is not the more diagnostic of the pair: on Lod-lille’s captured items mean head confidence is 0.631 against 0.295 on correct ones, the same inversion the top probability shows.

Answer types we did not use. The interface also accepts `score` (a bounded numeric judgement) and `boolean` questions. We use `choice` exclusively and deliberately. A `score` question would ask the model to rate a gloss in isolation, which destroys the property the instrument depends on: that the gold and the literal reading compete inside a single forward pass, exactly as the psycholinguistic account describes (Cacciari and Tabossi, 1988; Titone and Connine, 1999). A `boolean` question would do the same and additionally reintroduce a yes/no label whose name carries meaning of its own, which is the failure Sun et al. (2026) document. Keeping every experiment in one answer type also means the five distractor families differ in nothing but their option text.

Arm	Instruction string
Laya-ml	Bangla, see note
Lod-lille	Bangla, identical to above
Jev-1.13	“What is the actual figurative meaning of this idiom?”

Table 6: The instruction is one sentence and is byte-identical across families, experiments and the two Bangla arms. The Bangla string is the direct equivalent of the English one; it is not reproduced here because the typesetting toolchain used for this paper cannot shape Bengali script. It is released verbatim in `configs/task.json`.

B Positioning against concurrent work

The decision-model literature is a few weeks old at the time of writing, and several papers describe failures adjacent to ours. This appendix states what each one establishes, what it does not, and why our design is complementary rather than duplicative.

Mechanism versus behaviour. Oh et al. (2026) apply causal tracing to autoregressive transformers and recover two pathways carrying an idiom’s figurative and literal readings at once: early layers and

specific attention heads promote the figurative interpretation while a direct route keeps the literal one available, and disambiguating context is integrated from the first layer onward. That is mechanistic evidence that the competition our D-LIT condition exploits is real inside a model, and it is what makes our outcome measure interpretable rather than arbitrary: when a decision model places its mass on the literal gloss, there is a documented sense in which both readings were present and one won.

The two designs answer different questions, and the difference is the reason both are needed. Causal tracing requires weight access, token-level activations and a generative forward pass to intervene on. None of the three systems we audit offers the first, two of them emit no token stream to trace, and one is a commercial endpoint behind an HTTP interface. Their method cannot be pointed at the systems that are currently being deployed as routers and guardrails; ours can, at the cost of saying nothing about where inside the model the competition is resolved. They establish *how* the literal route survives; we measure *how often* it takes over, in which language, and with what confidence attached.

Option names and option definitions. Sun et al. (2026) hold a typed model’s option definitions fixed and permute which name is bound to which, and find decision-flip rates up to 70.4 points higher for semantically loaded labels than for neutral ones: the model follows the name rather than the definition it was given. They note that type-safety conceals this, because every output remains structurally valid.

Our O-series (§5.5) manipulates the complementary half. We hold the option names fixed at A–D and destroy the internal structure of the text bound to them, by truncation and by shuffling words within each option, and find accuracy almost unchanged. Read together, the two results bracket the same conclusion from either side: within an option, these models are sensitive to lexical identity and insensitive to composed meaning. Neither result alone would establish that, because theirs is consistent with a model that reads definitions carefully but weights names more heavily, and ours is consistent with a model that reads definitions as unordered bags.

Context that flips decisions. Xu (2026) append short, fluent, task-preserving context and redirect 61.4% of initially correct decisions, with targeted flip rates of 64.9–73.2% across systems. Our E3

carrier-sentence condition (§5.6) is the benign case of the same mechanism: embedding an idiom in its gold example sentence *raises* accuracy by 0.080 and 0.117 on the two open arms. The two findings are the same sensitivity with opposite signs, and the uncomfortable implication is that a caller cannot tell them apart from the interface. A model whose answer improves when helpful context is appended is, necessarily, a model whose answer moves when context is appended at all.

Calibration and cascades. Ibrahim and Zaki (2026) evaluate decision models against 19 language models on 18 computational social science tasks, and report a median deficit of 11.6 F1 at 44 times lower cost, with confidence calibration *exceeding* most baselines; they recommend decision models as a cheap first stage that routes uncertain items onward. Li et al. (2026) build that system, accepting high-confidence verdicts and escalating the rest, and exceed a frontier model’s accuracy at 41% of its cost.

We reproduce the favourable calibration for the commercial arm on its own language (§5.2), so this is not a contradiction. The boundary we add has two parts. First, in-distribution calibration does not survive a targeted distractor shift, and it fails in the direction that costs a cascade most: on the literal-bearing families the high-confidence tail is where the open arms are *least* accurate, so a confidence gate forwards exactly the items that should have been escalated. Second, and independent of any distribution assumption, the semantics-free control (§5.3) shows what each arm reports when the option list provably carries no information. A gate calibrated on real options has no defence against that case, and a degenerate option list is an ordinary consequence of an upstream bug or an empty retrieval.

What none of this settles. All of the above, ours included, is zero-shot evaluation of base checkpoints. Tang and Zheng (2026) review 28 papers published within nine days of the first widely used release and conclude that the typed readout has shown no independent accuracy advantage, with the demonstrated gains in latency, cost and cascade placement. Whether any of these failure modes survives task-specific fine-tuning is, as far as we can determine, unmeasured by anyone so far.

state sent to the model **ওজন** *ojon* · literally physical weight · English equivalent *carry weight*
instruction (identical for every item and every family): নিচের বাংলা বাগধারাটির প্রকৃত ভাবার্থ কোনটি?

option text, as supplied	role	Laya-ml p	Jev-1.13 p
A একগৰ্ভা; এক সন্তানের জননী; যে নারী একবার মাত্র গৰ্ভ ধারণ করে	distractor	0.00	0.02
B ওজন (physical weight)	literal	0.99	0.15
C ক্ষমতা, গুরুত্ব, মর্যাদা	gold	0.00	0.82
D হালকাতাবে পড়া	distractor	0.01	0.01

Figure 7: One item exactly as the two models received it, in the original Bengali script. The state is the idiom *ojon*; the four option texts are the real ones, reconstructed with the builder and seed the run used and checked against the recorded gold and literal keys. The probabilities are the returned values. Laya-ml places 0.99 on the **literal** option, which names the weight of an object, and 0.00 on the gold figurative gloss; Jev-1.13 places 0.82 on the gold. This figure was rendered outside $\text{\LaTeX}$ because $\text{\pdf\LaTeX}$ cannot shape Bengali; it is vector, with embedded font subsets and selectable text.

C What each experiment changes

Table 7 states, for each experiment, the one thing that varies and the things held fixed. Reading down the “varies” column follows the argument: each row isolates one candidate explanation for the failure measured in E1.

Varies	Fixed	Seed
E1 distractors	idiom, instr., k	1000
E2 idiom surface	meaning, options	2000
E3 idiom form	options, instr.	3000
E4 task language	item, family	4000
E5 nothing (derived)	n/a	n/a
E6 option text	idiom, gold	6000
E7 option order	option set	7000

Table 7: The seven interventions. E5 issues no calls of its own; it is computed from the E1 outputs. Seeds are the Bangla arms’; the commercial arm uses a disjoint set (Appendix Q) because its blocks were run in a different order.

E1, the ladder. D-RAND samples three other idioms’ figurative glosses uniformly. D-SURF samples glosses sharing at least one surface token with the gold. D-SEM samples glosses from idioms sharing at least one semantic tag. D-LIT replaces one sampled distractor with the idiom’s own expert literal gloss. D-HARD applies all three pressures at once. Duplicates are removed and the slate is topped up from the random pool if fewer than k distinct options survive.

E3, perturbing the unit. E3 runs on the literal-bearing option set, so its accuracies are comparable to D-LIT rather than to D-RAND: on the same 1,778 items Laya-ml scores 0.176 under D-LIT and 0.353 under D-RAND, against 0.158 for the intact

idiom here. v0 is the intact idiom. v1 shuffles its words, v2 reverses them, v3 deletes one word at random, v4 substitutes one word with a word sampled from another idiom, and v5 replaces the idiom with its gold carrier sentence. The option list is held byte-identical across all six, so the only difference reaching the model is the state string.

E6, ablating the options. O0 is the full gloss. O1 truncates every option to its first six words. O2 shuffles the words within each option. O3 replaces each gloss with its counterpart in the other language. O4 replaces every option with a bare label carrying no semantics, which pins accuracy at chance by construction and is the control of §5.3.

D Why interventions on the option set

This appendix explains the design choice the rest of the paper rests on, for readers who want the reasoning rather than the result.

A decision model takes a question and a list of candidate options and returns a probability over exactly those options. It is not autoregressive, it does not sample, and it emits no text. Three of the usual explainability routes are therefore closed. There is no rationale to inspect and no chain of thought whose faithfulness could be measured (Jacovi and Goldberg, 2020; Turpin et al., 2023). There are no weights for two of our three arms and none at all for the commercial one, which rules out probing, causal tracing and anything else that requires opening the model (Oh et al., 2026). And input attribution, the usual fallback (Ribeiro et al., 2016; Lundberg and Lee, 2017), is not merely unreliable here (Jain and Wallace, 2019; Adebayo et al., 2018;

Hooker et al., 2019) but ill-posed for the material.

The last point is worth stating carefully, because it is what motivates everything else. An idiom is non-compositional by definition (Nunberg et al., 1994): the meaning of the whole is not a function of the meanings of the parts. A Shapley value over the tokens of *kick the bucket* asks how much each word contributes to the figurative reading, and the answer is that the question has no referent. The words do not carry fractions of the meaning; the expression carries it as a unit. Any attribution method will still return numbers, and those numbers will look like an explanation, which is worse than returning nothing.

What remains is the interface itself. The caller controls the question and the option list and observes a probability vector. So we intervene there. The corpus supplies, for every idiom, both a gold figurative gloss and an expert literal gloss, which means a counterfactual distractor is already annotated and we create no new labels: dropping the literal gloss into the option list turns a benchmark item into a targeted probe of whether the expression was read as an idiom or as its words. Holding the item, the instruction and k fixed while varying only the distractor family makes the comparison within-item and within-model, so a difference between two families cannot be explained by item difficulty, by the prompt, or by anything about the model that does not change between the two calls.

The same logic produces the other six interventions. If the failure were about the idiom’s surface form rather than its meaning, E2 would show it. If the model held the expression as a stored unit, breaking the unit in E3 would hurt. If the problem were figurativeness rather than the language, E4 would not move. If the model read options as sentences rather than as bags of words, E6 would collapse under scrambling. If the scorer were order-independent as documented, E7 would return the same gloss under all 24 permutations. Each experiment is an attempt to explain the E1 result away, and the paper reports which attempts succeed.

E The full distractor ladder

Table 8 gives every family for every arm. Two caveats on the commercial column: its numbers are English, and its D-SURF and D-SEM rows run on the larger 8,761-item English pool rather than the 3,265-item literal pool, because those families need no literal gloss.

Family	Laya-ml bn	Lod-lille bn	Jev-1.13 en
D-RAND	0.367	0.484	0.968
D-SURF	0.338	0.412	0.915
D-SEM	0.333	0.446	0.829
D-LIT	0.151	0.066	0.931
D-HARD	0.144	0.063	0.854
LAR D-LIT	0.644	0.885	0.047
LAR D-HARD	0.630	0.855	0.043

Table 8: Accuracy by distractor family, and literal attraction on the two families that contain a literal gloss. Chance is 0.250. The commercial column is English and is the reason §5.1 runs the matrix: compare it with the Bangla column of Table 12, where the same model is captured at 0.304.

F Selective prediction in detail

Selective prediction asks what happens if the caller answers only the items the model is most confident about. A useful confidence signal makes accuracy rise as coverage falls. Table 9 reports accuracy at low coverage for the two open arms, on the random-distractor condition and on the literal-bearing one.

coverage	D-RAND	D-LIT	D-RAND	D-LIT
0.05	0.669	0.007	0.977	0.000
0.10	0.597	0.007	0.989	0.000
0.20	0.528	0.023	0.899	0.000
0.35	0.470	0.047	0.785	0.000
full	0.367	0.151	0.484	0.066

Table 9: Accuracy under a confidence gate. On random distractors both arms behave as a caller would hope: restricting to the most confident 5% roughly doubles accuracy. On the literal-bearing condition the curve inverts. Laya-ml falls from 0.151 at full coverage to 0.007 in its most confident 5%, and Lod-lille answers **every one** of its most confident 35% incorrectly. Gating on confidence is worse than not gating, and worse than gating at random.

This is the result with the most direct operational consequence. A cascade that forwards high-confidence answers and escalates the rest (Li et al., 2026; Ibrahim and Zaki, 2026) forwards precisely the items on which these models are wrong, and escalates the ones they would have got right.

G Stratified results

Literal capture is not concentrated in rare or obscure idioms, which would be the comfortable explanation. Splitting Laya-ml’s D-LIT condition by the corpus’s own frequency annotation gives LAR 0.671 on *very common* idioms ($n = 1,146$), 0.646 on *common* ($n = 6,200$), 0.615 on *rare* ($n = 1,507$) and 0.565 on *very rare* ($n = 23$).

If anything the effect is strongest on the most familiar expressions, which is the opposite of a long-tail story and consistent with these being the idioms whose literal readings are most available.

The corpus’s `cultural_significance` field is unusable as a stratifier: 10,337 of 10,359 rows carry the same value.

H Output distributions

The three arms do not merely differ in accuracy; they place their probability mass very differently, and the shape of that mass is what makes the confidence results of §5.2 possible. Table 10 summarises the returned vectors.

		mean	med.	>.95	ent.
Laya	D-RAND	0.438	0.408	0.001	0.892
	D-LIT	0.536	0.479	0.039	0.778
Lod	D-RAND	0.447	0.391	0.002	0.881
	D-LIT	0.745	0.819	0.171	0.497
Jev	D-RAND	0.967	1.000	0.852	0.068
	D-LIT	0.931	0.990	0.708	0.131

Table 10: Distribution of the returned maximum probability, and normalised entropy of the full vector (1 = uniform). The literal-bearing family makes both open arms *sharper*, not flatter: Lod-lille’s mean peak rises from 0.447 to 0.745 and the share of near-deterministic answers from 0.002 to 0.171, while its accuracy falls from 0.484 to 0.066. The model is not confused by the literal gloss; it is convinced by it.

That last point is worth stating separately, because it is the mechanism behind the inverted reliability curves. A model that found the literal option merely distracting would spread mass and lose accuracy together. Both open arms do the opposite: adding the literal gloss *concentrates* the distribution onto it. On 24 items Laya-ml returns $p = 1.000$ on the literal gloss and $p = 0.000$ on the gold.

Worked examples. Table 11 gives four of them. The idioms are transliterated because the toolchain cannot set Bengali script; the released data carries the original strings.

Idiom	Literal	Figurative (gold)	p_{lit}
tel	oil	arrogance	1.000
chamcha	spoon	sycophant	1.000
til	sesame seed	a tiny amount	1.000
charpaya	four-legged	rope cot	1.000

Table 11: Laya-ml under D-LIT. In each case the expert literal gloss and the expert figurative gloss are both in the option list, and the model places all of its mass on the literal reading. These are not hard or obscure idioms; three of the four are single words in everyday use.

	Bangla		English	
	acc	LAR	acc	LAR
Laya-ml	0.127	0.731	0.417	0.379
Lod-lille	0.044	0.918	0.684	0.241
Jev-1.13	0.536	0.304	0.926	0.047

Table 12: The literal condition as a model $\times$ language matrix, all six cells measured on the same 3,265 items with one construction and one seed ($n = 3,265$ per cell; 95% CIs in Appendix I). Bold marks the highest LAR per language. Every arm is far more captured in Bangla than in English, and the Bangla ordering does not follow capability: Jev-1.13 is the most accurate and the least captured, while Lod-lille is more accurate than Laya-ml and more captured.

I The capture matrix

Table 12 gives the six cells of §5.1 with the accuracy column alongside LAR. All cells are $n = 3,265$ and were built by one function from seed 21000; 95% bootstrap intervals on LAR are [.716, .746], [.363, .397], [.909, .928], [.227, .256], [.288, .319] and [.040, .055] reading down the table. The paired within-model language effect on capture is +0.352 for Laya-ml, +0.677 for Lod-lille and +0.256 for Jev-1.13, all McNemar $p < 10^{-100}$. Jev-1.13’s English cell reproduces the value reported from the original run, 0.0472, which is why we treat the two constructions as measuring the same thing.

J Cross-lingual arms

Figure 10 plots the E4 conditions discussed in §5.7.

K The matched three-way condition

Figure 11 plots Table 3, the one condition that is identical across all three arms.

L Unit integrity

Figure 12 plots the E3 perturbations described in §5.6. The options are byte-identical across all six variants, so the only difference reaching the model is the state string.

M Surface invariance

The corpus ships alternative surface forms certified by annotators as meaning-preserving. A model reading meaning should be invariant across such a pair. Laya-ml agrees with itself on 0.657 of 3,143 pairs (2,423 items) and Lod-lille on 0.677 of 310 pairs (234 items): about a third of the time, a meaning-preserving rewrite changes the answer.

what each manipulation does to real Bangla text idiom অগাধ জলের মাছ, English equivalent <i>Still waters run deep</i>		
E3, the state changes	string sent as the state	p(gold)
intact	অগাধ জলের মাছ	0.21
word shuffle	মাছ জলের অগাধ	0.19
reversed	মাছ জলের অগাধ	0.19
delete one word	অগাধ জলের	0.24
substitute one word	অগাধ জলের ফ্যা	0.24
in its carrier sentence	সে অগাধ জলের মাছ, কখনো বুঝতে পারি না তার পরিকল্পনা কী।	0.07
E6, the options change	the gold option, rewritten; the other three are rewritten the same way	
full gloss	খুব চালাক লোক; সুচতুর ব্যক্তি; যার মনোভাব বোঝা যায় না	
truncated to 6 words	খুব চালাক লোক; সুচতুর ব্যক্তি; যার	
words scrambled	যায় সুচতুর চালাক ব্যক্তি; বোঝা খুব না মনোভাব লোক; যার	
English gloss	A very clever person; a shrewd individual; someone whose mindset or intentions are hard to fathom	
semantics removed	A	

Figure 8: What the interventions do to real Bangla text, for the idiom *agadh joler machh* (*still waters run deep*, literally “a fish of deep water”). *Top*: the six E3 states, which are the exact strings logged during the run, with the probability the model gave the gold gloss. Shuffling and reversing a non-compositional expression leaves that probability essentially unchanged, and only the carrier sentence moves it. *Bottom*: the five E6 option forms, applied here to the gold option; the other three options are rewritten the same way. Truncation and word-scrambling preserve the lexical content that the scoring appears to depend on, while the last row removes it entirely and is the control of §5.3.

Mean Jensen–Shannon divergence between the two probability vectors is 0.039 and 0.085.

N Subset representativeness

Lod-lille ran a fixed 10% random subset. Table 13 compares Laya-ml’s accuracy on those same 888 items against its own full 8,876.

Family	full	subset	drift
D-RAND	0.367	0.351	−0.016
D-SURF	0.338	0.339	+0.001
D-SEM	0.333	0.325	−0.007
D-LIT	0.151	0.136	−0.015
D-HARD	0.144	0.149	+0.004

Table 13: Laya-ml on the 10% subset against its own full set. Maximum drift 0.016. Two of five subset estimates fall marginally outside the full-set bootstrap CI, which is expected at $n = 888$ and changes no sign or ordering. Cite the full-set figures as the precise ones.

O Full option ablation

Table 14 gives the numbers behind Figure 13.

P Order sensitivity and position bias

Figure 14 gives the per-slot selection shares behind the summary in §5.4.

Option text	Laya	Lod	Jev
full gloss	0.365	0.492	0.970
truncated	0.361	0.468	0.958
scrambled	0.357	0.479	0.974
other lang.	0.347	0.398	0.963
no semantics	0.256	0.233	0.254

Table 14: Option-side ablation. The grey row is the semantics-free control, at chance by construction in all three arms. Jev-1.13’s column is English.

Q Run record

Environment. Python 3.13.15 throughout. The two open arms ran on one Tesla T4 (15,360 MiB): Laya-ml in bf16 with `max_len` 1024 at 27.2 ms per call, Lod-lille in fp32 at 86.1 ms per call. The commercial arm needs no accelerator and ran on CPU against `typesafe/jev-1.13`.

Temperatures. Laya-ml’s run recorded `temperature = [1.0, 1.0, 1.0]` with `temperature_by_options` empty, so the per-bucket temperatures it ships were not applied and its ECE figures are raw. Lod-lille ships `temperature: 1.0` by design, its authors having reported that fitting hurt held-out calibration. Both open arms are therefore untempered and comparable on that axis. The commercial arm’s internal calibration is not observable.

items where Laya-ml places its mass on the literal gloss

Both glosses below were in the option list. The last column is the probability the model assigned to the literal one.

idiom	expert literal gloss	gold figurative gloss	English equivalent	p(literal)
ঈশ্বরপ্রাপ্তি	ঈশ্বরকে প্রাপ্তি (attainment of God)	মৃত্যু	<i>kick the bucket</i>	0.97
আকাশের চাঁদ	আকাশের চাঁদ (sky's moon)	অসম্ভব বা দুর্লভ বস্তু; বহু কাঙ্ক্ষিত বস্তু	<i>Pie in the sky</i>	0.99
অল্প জলের মাছ	অল্প জলের মাছ (little water's fish)	সামান্য পুঁজির বা সম্পদের মালিক; অল্প ধন নিয়ে গর্বকারী ব্যক্তি	<i>Big frog in a small puddle</i>	0.97
ওজন	ওজন (physical weight)	ক্ষমতা, গুরুত্ব, মর্যাদা	<i>carry weight</i>	0.99
অগ্নিপরীক্ষা	অগ্নির পরীক্ষা (fire test)	কঠিন পরীক্ষা; কঠিন পরিস্থিতির সম্মুখীন	<i>trial by fire</i>	0.96

Figure 9: Five items on which Laya-ml assigns almost all of its probability to the idiom’s own literal gloss, shown in Bengali with the English equivalent the annotators recorded. These are not obscure expressions: *ishwarprapti* is the Bangla *kick the bucket* and is read as “attainment of God”, *akashar chand* is *pie in the sky* and is read as “the moon of the sky”. Both glosses were present in the option list, and the right-hand column is the mass placed on the literal one.

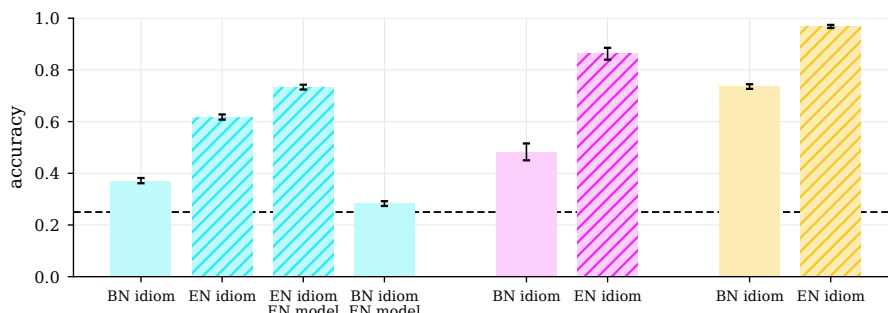


Figure 10: Accuracy by task language, grouped by model: Laya-ml (four bars), Lod-lille, Jev-1.13. Solid bars are Bangla input, hatched bars English; Laya-ml’s third and fourth bars use the English-only checkpoint, the fourth being the control. Error bars 95% bootstrap CIs, dashed line chance. Every model gains 20–45 points from the same idiom presented in English.

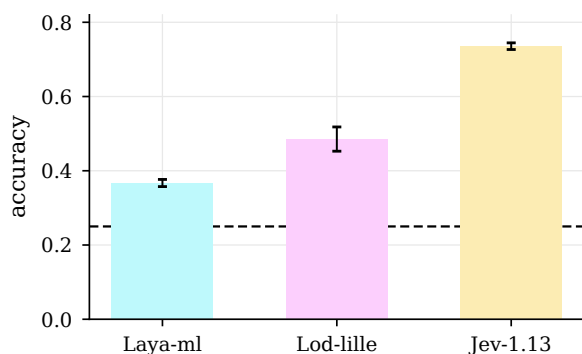


Figure 11: The matched three-way comparison. Bars left to right: Laya-ml, Lod-lille, Jev-1.13. Black dashed line is chance, error bars 95% bootstrap CIs. This capability ordering is the baseline against which the calibration failures in §5.2–§5.3 should be read.

Seeds. The Bangla arms use a per-experiment seed (E1 1000, E2 2000, E3 3000, E4 4000, E6 6000, E7 7000) added to the item id, so each item’s option permutation is reproducible and identical

across arms. The split seed is 20261001. The commercial arm uses a disjoint block seed set (order audit 17000, literal 12000, ceiling 11000, Bengali control 14000, options 16000) because its blocks ran in a different order; the permutation for a given item is still a pure function of seed and id.

Call volume and cost. The commercial arm issued 56,533 calls for \$0.9348 at \$0.042 per million input tokens with output free. The blocks are the order audit ($300 \times 24 = 7,200$), literal capture on the shared pool ($3,265 \times 3 = 9,795$), the English ceiling ($8,761 \times 3 = 26,283$), the Bengali control (8,761) and the option ablation ($873 \times 5 = 4,365$), which is 56,404 retained rows. The remaining 129 calls are the endpoint probe and the response-contract checks made before each block. Transient 429 and 5xx responses were retried with exponential backoff to a 30 s ceiling; no call was dropped.

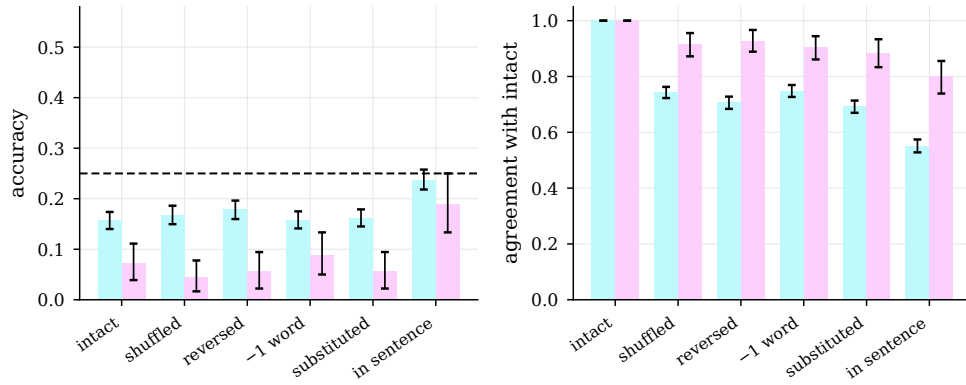


Figure 12: Unit integrity: the idiom is perturbed, the options held fixed. ■ Laya-ml, ■ Lod-lille; error bars 95% bootstrap CIs, dashed line chance (left panel). *Left*: accuracy. *Right*: agreement with the answer on the intact idiom. Scrambling and reversing a non-compositional expression does not meaningfully hurt either model, which is the bag-of-words signature.

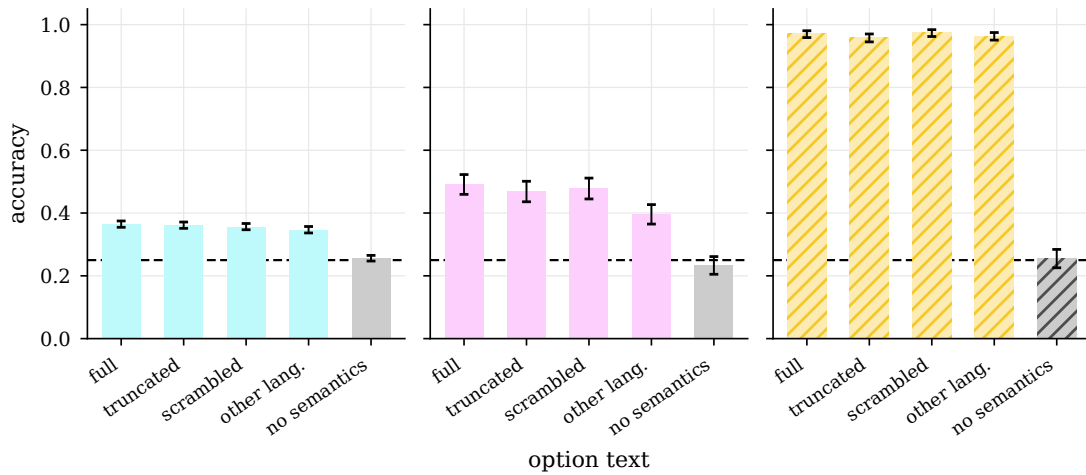


Figure 13: Option-side ablation. Panels left to right: ■ Laya-ml, ■ Lod-lille, ■ Jev-1.13. In each panel the fifth, ■ bar is the semantics-free control of §5.3; the whole Jev-1.13 panel is hatched because its task is in **English**. Error bars are 95% bootstrap CIs and the black dashed line is chance. Truncating and word-scrambling the option text cost almost nothing in any of the three models.

Statistics. All intervals are 95% percentile bootstrap over items, 2,000 resamples, fixed seed 0. ECE uses 15 equal-width bins. Paired comparisons use McNemar’s exact test via the binomial distribution. AURC is reported against a random-gate baseline computed on the same items.

What is released. The per-item outputs for every cell of every experiment (one row per item per condition, carrying the full probability vector, the predicted key, the gold key, the literal key and the gold position), the five split manifests, the three run environment records quoted above, the task configurations containing the verbatim instruction strings, the analysis scripts that turn those outputs into every table in this paper, the figure sources, and a CSV of the exact plotted values beside each figure.

Pairing. Because the two open arms were reconstructed independently from a written contract rather than from shared code, the released outputs admit a direct check: on the 4,440 overlapping rows the gold key, gold position, idiom string and literal-distractor key agree at 1.000. That check is itself a released artifact, and it is what licenses every paired test in the paper.

The matrix run. The six cells of §5.1 were run afterwards as one job, with its own seed namespace (21000) so no option permutation collides with the runs above. Every cell covers the same 3,265 items and is built by a single function; the two languages share one distractor draw per item and are shuffled independently from the same generator. Laya-ml ran both languages on the multilingual checkpoint, so only the language changes and the checkpoint does not. Lod-lille ran the full pool rather than its earlier 10% subset, which supersedes the thin

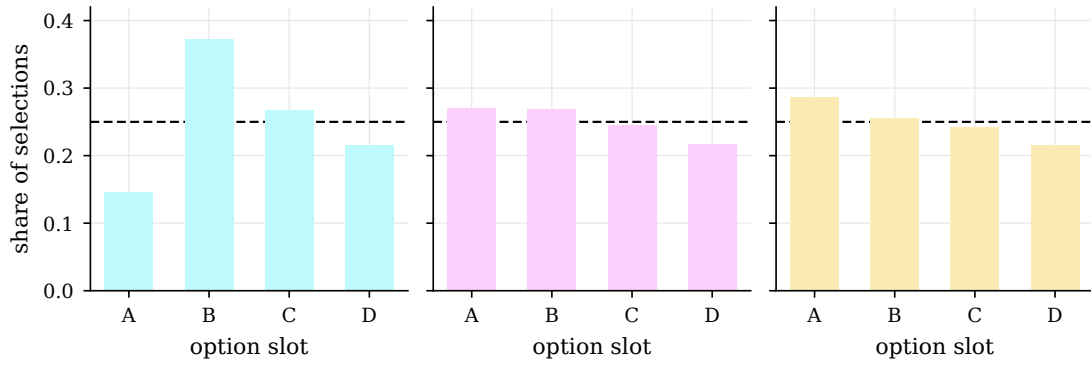


Figure 14: Share of selections falling on each option slot. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. The gold option is placed uniformly at random, so the black dashed line at 0.25 is the expectation under no positional bias and any departure from it is bias. Laya-ml takes slot B 37.2% of the time and slot A 14.6%; the other two show only a mild primacy gradient.

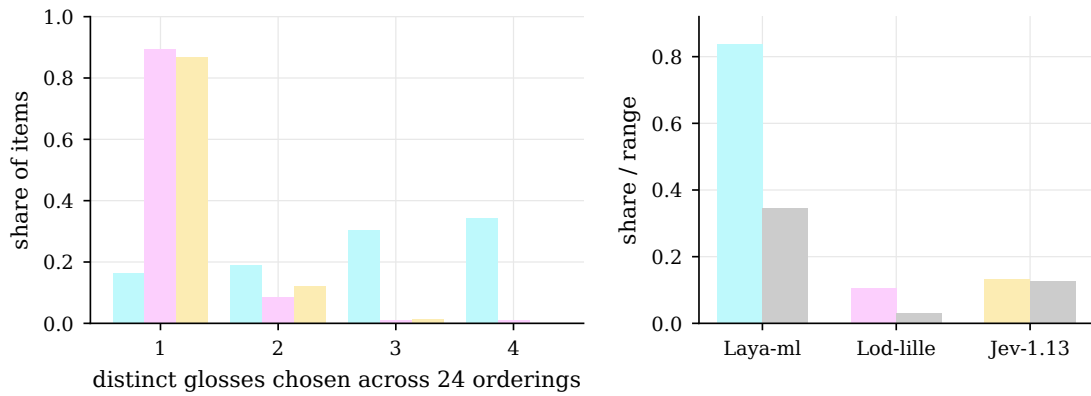


Figure 15: Order sensitivity. *Left*: how many distinct glosses each model selects across the 24 orderings of one option set; within each group bars run Laya-ml, Lod-lille, Jev-1.13, and a value of 1 means perfect stability. *Right*: per model, the coloured bar is the share of items whose chosen gloss is unstable and the grey bar is the mean within-item range of $p(\text{gold})$. Laya-ml is unstable on 83.7% of items with a mean $p(\text{gold})$ swing of 0.346; the other two are near-stable.

316-item in-pool estimate. The commercial arm issued 6,531 calls for \$0.1152. Its English cell reproduces the value from the original run to four decimal places (0.0472), which is the check that the two constructions measure the same thing.

Known gaps. E2 was not run on the commercial arm, because the alternative surface forms exist only in Bangla and the arm's task is English. The matrix covers the literal condition only; the other distractor families were not run in both languages. The confidence and calibration results remain two-arm, because the commercial arm's reliability was measured on its English run.